

Pareto Optimality Is Incompatible with a Bounded Per-Agent Subsidy: A Minimal Example

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Abstract

We exhibit a two-agent, three-chore instance with binary marginal costs in which *no* complete allocation is simultaneously (i) envy-free once each agent is paid a subsidy of at most one unit, and (ii) Pareto optimal. The two properties are not merely hard to achieve together on this instance — they are mutually exclusive: the set of allocations satisfying the subsidy guarantee and the set of Pareto optimal allocations are disjoint. We use the standard definition of Pareto optimality of Chakraborty, Igarashi, Suksompong and Zick [1], translated from goods to chores by the usual sign flip, and we spell out one point where the two settings genuinely differ — allocations must be taken complete for the chores version of the definition to be non-vacuous.

1 Summary, for the impatient reader

- Two agents share the cost function $c(S) = \min(|S|, 2)$ on three chores: the first two chores hurt, the third is free once you already hold two.
- Concentrating all three chores on one agent is the *cheapest* outcome in total (cost 2), because the third chore is absorbed for free. Splitting the chores always costs 3 in total.
- But concentration creates an envy gap of exactly 2: the loaded agent needs a subsidy of 2 to stop envying the agent holding nothing. A guarantee capping the subsidy at 1 per agent forbids this.
- Every allocation that respects the cap is a split, and every split is Pareto dominated by concentrating the chores.
- Conclusion: on this instance, **no allocation is both Pareto optimal and feasible under the $\{0, 1\}$ -subsidy guarantee**. This is not an artifact of a bad tie-break; the two requirements are logically incompatible here.

2 The setting

There are $N = \{1, \dots, n\}$ agents and M a finite set of indivisible *chores*. Each agent i has a cost function $c_i : 2^M \rightarrow \mathbb{Z}_{\geq 0}$ with $c_i(\emptyset) = 0$. We say c_i has *binary marginals* if

$$c_i(S \cup \{e\}) - c_i(S) \in \{0, 1\} \quad \text{for every } S \subseteq M \text{ and every } e \notin S.$$

No further structure is assumed: c_i need not be additive, submodular, or subadditive, and different agents' cost functions need not be related.

A *complete allocation* $A = (A_1, \dots, A_n)$ partitions M among the agents. A subsidy vector $p \in \mathbb{R}_{\geq 0}^n$ makes (A, p) *envy-free* if

$$c_i(A_i) - p_i \leq c_i(A_j) - p_j \quad \text{for all } i, j \in N.$$

We are interested in the following guarantee, which caps the subsidy paid to any single agent:

Definition 2.1 (The $\{0, 1\}$ -subsidy property). *A complete allocation A is S -feasible if there exists $p \in \{0, 1\}^n$ making (A, p) envy-free.*

Definition 2.1 is the property that a subsidy of at most one unit per agent suffices to eliminate all envy under A . This note asks whether an S -feasible allocation can always be chosen to be Pareto optimal as well.

3 Pareto optimality

3.1 The standard definition, for goods

We use the definition of Chakraborty, Igarashi, Suksompong and Zick [1, Sec. 2], stated for their setting of *goods*, where each agent has a value function $v_i : 2^M \rightarrow \mathbb{R}_{\geq 0}$ (higher is better):

An allocation A' is said to Pareto dominate an allocation A if $v_i(A'_i) \geq v_i(A_i)$ for all agents $i \in N$ and $v_j(A'_j) > v_j(A_j)$ for some agent $j \in N$. An allocation is Pareto optimal (or PO for short) if it is not Pareto dominated by any other allocation.

3.2 The chores version

Our setting is *chores*, so we work with costs rather than values. The standard translation between the two is $v_i = -c_i$: a chore is something an agent does not want, so cost is disutility. Substituting $v_i = -c_i$ into the inequalities above,

$$v_i(A'_i) \geq v_i(A_i) \iff -c_i(A'_i) \geq -c_i(A_i) \iff c_i(A'_i) \leq c_i(A_i),$$

and the strict inequality flips the same way. The chores form of the definition is therefore *the same statement*, not an independent one:

Definition 3.1 (Pareto domination, chores form). *Allocation B Pareto dominates allocation A if $c_i(B_i) \leq c_i(A_i)$ for every $i \in N$, with strict inequality for at least one $j \in N$. An allocation is Pareto optimal (PO) if no allocation Pareto dominates it.*

3.3 Why allocations must be taken complete

Chakraborty et al. allow *incomplete* allocations — some items may go unassigned — and Pareto optimality is defined over all allocations, complete or not. For goods this costs nothing: withholding a good from every agent can only make everyone worse off (assuming positive value), so an incomplete allocation is automatically dominated by completing it, and hence can never be PO. Chakraborty et al. note exactly this in their proof of Theorem 4.2.

For chores the same argument runs backward. Withholding a chore makes *every* agent weakly better off. If incomplete allocations were allowed to compete as dominators, *every* complete allocation would be dominated by the empty allocation (assign nothing to anybody), and the notion

of Pareto optimality would collapse to nothing being PO except the empty allocation itself — a vacuous statement that says nothing about how to divide the chores that must, in fact, be divided.

We therefore adopt the standard convention of the chore fair-division literature and restrict Definition 3.1 to *complete* allocations throughout: an allocation is a partition of M , and domination is checked only against other partitions of M . This is not a weaker or different notion of Pareto optimality; it is the same definition, applied to the domain on which it is non-vacuous for chores.

4 The instance

Example 4.1. Let $n = 2$ agents and $M = \{a, b, c\}$, three chores. Both agents share the same cost function

$$c(S) = \min(|S|, 2).$$

$ S $	0	1	2	3
$c(S)$	0	1	2	2

Table 1: The shared cost function. The first two chores cost one unit each; the third is free once the agent already holds two.

Proposition 4.2. The cost function of Example 4.1 has binary marginals, is monotone, satisfies $c(\emptyset) = 0$, and is not additive.

Proof. $c(\emptyset) = \min(0, 2) = 0$. Monotonicity is immediate from the definition of $\min$. For the marginals: adding a chore to the empty bundle gives $c(\{e\}) - c(\emptyset) = 1 - 0 = 1$; adding a chore to a singleton gives $c(S) - c(\{e'\}) = 2 - 1 = 1$; adding a chore to a bundle of size two gives $c(M) - c(S) = 2 - 2 = 0$. Every marginal is in $\{0, 1\}$.

If c were additive, we would need $c(\{a, b, c\}) = c(\{a\}) + c(\{b\}) + c(\{c\}) = 1 + 1 + 1 = 3$. But $c(\{a, b, c\}) = \min(3, 2) = 2 \neq 3$, so c is not additive. $\square$

5 The eight allocations

With $n = 2$ and $|M| = 3$, there are $2^3 = 8$ complete allocations. Because both agents share the same cost function, the envy-free-with-subsidy condition simplifies sharply.

Lemma 5.1 (Forced subsidy gap). For $n = 2$ agents with a shared cost function c , an allocation A admits an envy-free subsidy vector p if and only if

$$p_1 - p_2 = c(A_1) - c(A_2).$$

Proof. Envy-freeness requires both $c(A_1) - p_1 \leq c(A_2) - p_2$ (agent 1 does not envy agent 2) and $c(A_2) - p_2 \leq c(A_1) - p_1$ (agent 2 does not envy agent 1). Rearranging, the first gives $p_1 - p_2 \geq c(A_1) - c(A_2)$ and the second gives $p_1 - p_2 \leq c(A_1) - c(A_2)$. Together these force equality. $\square$

Lemma 5.1 leaves no freedom at all: once A is fixed, the difference $p_1 - p_2$ is determined exactly. Since $p \in \{0, 1\}^2$, the achievable differences are only $\{-1, 0, 1\}$; the allocation is S -feasible (Definition 2.1) exactly when $|c(A_1) - c(A_2)| \leq 1$.

A_1	A_2	$(c(A_1), c(A_2))$	$p_1 - p_2$ forced	S -feasible?	PO?	dominated by
$\emptyset$	$\{a, b, c\}$	$(0, 2)$	-2	no	yes	—
$\{a\}$	$\{b, c\}$	$(1, 2)$	-1	yes	no	$(\emptyset, \{a, b, c\})$
$\{b\}$	$\{a, c\}$	$(1, 2)$	-1	yes	no	$(\emptyset, \{a, b, c\})$
$\{c\}$	$\{a, b\}$	$(1, 2)$	-1	yes	no	$(\emptyset, \{a, b, c\})$
$\{a, b\}$	$\{c\}$	$(2, 1)$	1	yes	no	$(\{a, b, c\}, \emptyset)$
$\{a, c\}$	$\{b\}$	$(2, 1)$	1	yes	no	$(\{a, b, c\}, \emptyset)$
$\{b, c\}$	$\{a\}$	$(2, 1)$	1	yes	no	$(\{a, b, c\}, \emptyset)$
$\{a, b, c\}$	$\emptyset$	$(2, 0)$	2	no	yes	—

Table 2: All eight complete allocations of Example 4.1. Exactly two are Pareto optimal (the concentrated ones); exactly six are S -feasible (the splits). The two sets do not overlap.

6 The two Pareto optimal allocations

Proposition 6.1. *The allocations $(\{a, b, c\}, \emptyset)$ and $(\emptyset, \{a, b, c\})$ are Pareto optimal, and no other allocation is.*

Proof. Let $A = (\{a, b, c\}, \emptyset)$, with costs $(2, 0)$. Suppose B dominates A : then $c(B_1) \leq 2$ and $c(B_2) \leq 0$, with at least one strict. Costs are non-negative, so $c(B_2) \leq 0$ forces $c(B_2) = 0$, and the only subset with cost 0 is $\emptyset$ (Table 1), so $B_2 = \emptyset$. Completeness then forces $B_1 = \{a, b, c\}$. So $B = A$: no allocation other than A itself satisfies the domination inequalities, and A is PO. The argument for $(\emptyset, \{a, b, c\})$ is symmetric.

For every other allocation, Table 2 exhibits an explicit dominator. Take $(\{a, b\}, \{c\})$, with costs $(2, 1)$, as a representative case: $c(\{a, b\}) = 2 = c(\{a, b, c\})$, while $c(\{c\}) = 1 > 0 = c(\emptyset)$. So $(\{a, b, c\}, \emptyset)$ dominates it — agent 1 is unaffected because her bundle is already saturated, and agent 2 strictly improves. The same pattern verifies every other row. Hence no allocation outside $\{(\{a, b, c\}, \emptyset), (\emptyset, \{a, b, c\})\}$ is PO. $\square$

Remark 6.2 (The two PO allocations do not dominate each other). *It is worth checking directly that $(\{a, b, c\}, \emptyset)$, with costs $(2, 0)$, does not Pareto dominate $(\emptyset, \{a, b, c\})$, with costs $(0, 2)$ — and vice versa. Domination of $(0, 2)$ by $(2, 0)$ would require $2 \leq 0$ (agent 1’s cost) and $0 \leq 2$ (agent 2’s cost); the first fails. Domination in the other direction would require $0 \leq 2$ and $2 \leq 0$; again the second fails. So neither dominates the other. This is not a contradiction: Pareto optimality is a partial order, not a ranking, and $(2, 0)$ and $(0, 2)$ are simply two incomparable points on the Pareto frontier — moving from one to the other helps one agent while hurting the other, which is a trade-off, not an improvement. Both are undominated by anything, including each other, which is exactly what makes both of them Pareto optimal simultaneously.*

7 The incompatibility

Corollary 7.1 (Main observation). *On the instance of Example 4.1, no complete allocation is both S -feasible (Definition 2.1) and Pareto optimal (Definition 3.1).*

Proof. By Proposition 6.1, the Pareto optimal allocations are exactly $(\{a, b, c\}, \emptyset)$ and $(\emptyset, \{a, b, c\})$, with forced subsidy gaps $p_1 - p_2 = \pm 2$ (Lemma 5.1 and Table 2). Neither gap is achievable by any $p \in \{0, 1\}^2$, whose achievable gaps are $\{-1, 0, 1\}$. So neither PO allocation is S -feasible. Every

other allocation is S -feasible (Table 2) but, by Proposition 6.1, is not PO. Hence the two sets — S -feasible allocations and Pareto optimal allocations — are disjoint. $\square$

Remark 7.2 (Why: efficiency and fairness pull in opposite directions). *The two facts behind Corollary 7.1 are two sides of the same feature of c . Because c saturates at $|S| = 2$, the third chore is free to an agent who already holds two: moving it onto that agent costs nothing and relieves whoever else was holding it, which is exactly what makes concentration efficient (Pareto optimal). But that same saturation means the loaded agent’s cost jumps straight from 0 to 2 relative to the empty agent, an envy gap the $\{0, 1\}$ -subsidy guarantee cannot fund. The guarantee therefore forces a split; every split leaves one chore un-absorbed that someone else would have taken for free, which is exactly the source of the domination in Table 2.*

Remark 7.3 (The efficiency cost of the guarantee). *The cheapest Pareto optimal allocation has total cost $c(A_1) + c(A_2) = 2$. The cheapest S -feasible allocation has total cost 3 (every split costs $2 + 1$ in some order). Enforcing the $\{0, 1\}$ -subsidy guarantee therefore costs society one extra unit of total disutility on this instance — a 50% overhead relative to the efficient outcome.*

8 Further observations (informal)

The following two points were checked computationally and are recorded here for context; they are not needed for Corollary 7.1, which is already fully self-contained on the three-chore instance above.

- **The phenomenon is not an artifact of $n = 2, m = 3$.** The same construction, $c(S) = \min(|S|, t)$ with all agents sharing c , produces analogous instances — no S -feasible allocation is Pareto optimal — for a range of (n, m, t) with $t = 2$ and $m \geq n + 1$, checked exhaustively for $n \in \{2, 3, 4\}$.
- **Additivity seems to be exactly what is missing.** In every instance checked (exhaustive search at small sizes, and random sampling at larger sizes) where every agent’s cost function is additive, an S -feasible allocation that is also Pareto optimal was always found. Proposition 4.2 confirms that the instance above is emphatically not additive; that non-additivity appears to be essential to the incompatibility.

References

- [1] Mithun Chakraborty, Ayumi Igarashi, Warut Suksompong, and Yair Zick. *Weighted Envy-Freeness in Indivisible Item Allocation*. ACM Transactions on Economics and Computation, 9(3):1–18, 2021.